รหัสคำสั่งนีโมนิค ของ MCS-51 ตามลำดับตัวอักษร

MNEMONIC	OPERANDS	OPCODE	Cycles
ACALL	addr11	11 aaa	2
		31 aaa	2
		51 aaa	2
		71 aaa	2
		91 aaa	2
		B1 aaa	2
		D1 aaa	2
		F1 aaa	2
ADD	A,#data	24 nn	1
ADD	A,@R0	26	1
ADD	A,@R1	27	1
ADD	A,addr8	25 aa	1
ADD	A,R0	28	1
ADD	A,R1	29	1
ADD	A,R2	2A	1
ADD	A,R3	2B	1
ADD	A,R4	2C	1
ADD	A,R5	2D	1
ADD	A,R6	2E	1
ADD	A,R7	2F	1
ADDC	A,#data	34 nn	1
ADDC	A,@R0	36	1
ADDC	A,@R1	37	1
ADDC	A,addr8	35 aa	1
ADDC	A,R0	38	1
ADDC	A,R1	39	1
ADDC	A,R2	3A	1
ADDC	A,R3	3B	1
ADDC	A,R4	3C	1
ADDC	A,R5	3D	1
ADDC	A,R6	3E	1
ADDC	A,R7	3F	1
AJMP	addr11	01 aaa	2
		21 aaa	2
		41 aaa	2
		61 aaa	2
		81 aaa	2
		A1 aaa	2
		C1 aaa	2
		E1 aaa	2
ANL	A,#data	54 nn	1
ANL	A,@R0	56	1
ANL	A,@R1	57	1
ANL	A,addr8	55 aa	1
ANL	A,R0	58	1
ANL	A,R1	59	1
ANL	A,R2	5A	1
ANL	A,R3	5B	1
ANL	A,R4	5C	1
ANL	A,R5	5D	1
ANL	A,R6	5E	1
ANL	A,R7	5F	1
ANL	C,bit	B0 bb	2
ANL	C,/bit	82 bb	2
ANL	addr8,#data	53 aa nn	2
ANL	addr8,A	52 aa	1

MNEMONIC	OPERANDS	OPCODE	Cycles
CJNE	@R0,#data,rel	B6 nn rr	2
CJNE	@R0,#data,rel	B7 nn rr	2
CJNE	A,#data,rel	B4 nn rr	2
CJNE	A,addr8,rel	B5 dd rr	2
CJNE	R0,#data,rel	B8 nn rr	2
CJNE	R1,#data,rel	B9 nn rr	2
CJNE	R2,#data,rel	BA nn rr	2
CJNE	R3,#data,rel	BB nn rr	2
CJNE	R4,#data,rel	BC nn rr	2
CJNE	R5,#data,rel	BD nn rr	2
CJNE	R6,#data,rel	BE nn rr	2
CJNE	R7,#data,rel	BF nn rr	2
CLR	A	E4	1
CLR	bit	C2 bb	1
CLR	C	C3	1
CPL	A	F4	1
CPL	bit	B2 bb	1
CPL	C	B3	1
DA	A	D4	1
DEC	@R0	16	1
DEC	@R1	17	1
DEC	A	14	1
DEC	addr8	15 aa	1
DEC	R0	18	1
DEC	R1	19	1
DEC	R2	1A	1
DEC	R3	1B	1
DEC	R4	1C	1
DEC	R5	1D	1
DEC	R6	1E	1
DEC	R7	1F	1
DIV	AB	84	4
DJNZ	addr8,rel	D5 aa rr	2
DJNZ	R0,rel	D8 rr	2
DJNZ	R1,rel	D9 rr	2
DJNZ	R2,rel	DA rr	2
DJNZ	R3,rel	DB rr	2
DJNZ	R4,rel	DC rr	2
DJNZ	R5,rel	DD rr	2
DJNZ	R6,rel	DE rr	2
DJNZ	R7,rel	DF rr	2
INC	@R0	06	1
INC	@R1	07	1
INC	A	04	1
INC	addr8	05 aa	1
INC	DPTR	A3	2
INC	R0	08	1
INC	R1	09	1
INC	R2	0A	1
INC	R3	0B	1
INC	R4	0C	1
INC	R5	0D	1
INC	R6	0E	1
INC	R7	OF	1

MNEMONIC	OPERANDS	OPCODE	Cycles
JB	bit,rel	20 bb rr	2
JBC	bit,rel	10 bb rr	2
JC	rel	40 rr	2
JMP	@A+DPTR	73	2
JNB	bit,rel	30 bb rr	2
JNC	rel	50 rr	2
JNZ	rel	70 rr	2
JZ	rel	60 rr	2
LCALL	addr16	12 aaaa	2
LJMP	addr16	02 aaaa	2
MOV	@R0,#data	76 nn	1
MOV	@R0,A	F6	1
MOV	@R0,addr8	A7 aa	2
MOV	@R1,#data	77 nn	1
MOV	@R1,A	F7	1
MOV	@R1,addr8	A7 aa	2
MOV	A,#data	74 nn	1
MOV	A,@R0	E6	1
MOV	A,@R1	E7	1
MOV	A,addr8	E5 aa	1
MOV	A,R0	E8	1
MOV	A,R1	E9	1
MOV	A,R2	EA	1
MOV	A,R3	EB	1
MOV	A,R4	EC	1
MOV	A,R5	ED	1
MOV	A,R6	EE	1
MOV	A,R7	EF	1
MOV	bit,C	92 bb	2
MOV	C,bit	A2 bb	1
MOV	addr8,#data	75 aa nn	2
MOV	addr8,@R0	86 aa	2
MOV	addr8,@R1	87 aa	2
MOV	addr8,A	F5 aa	1
MOV	addr8,addr8	85 aa aa	2
MOV	addr8,R0	88 aa	2
MOV	addr8,R1	89 aa	2
MOV	addr8,R2	8A aa	2
MOV	addr8,R3	8B aa	2
MOV	addr8,R4	8C aa	2
MOV	addr8,R5	8D aa	2
MOV	addr8,R6	8E aa	2
MOV	addr8,R7	8F aa	2
MOV	DPTR,#data	90 nnnn	2
MOV	R0,#data	78 nn	1
MOV	R1,#data	79 nn	1
MOV	R2,#data	7A nn	1
MOV	R3,#data	7B nn	1
MOV	R4,#data	7C nn	1
MOV	R5,#data	7D nn	1
MOV	R6,#data	7E nn	1
MOV	R7,#data	7F nn	1
MOV	R0,A	F8	1
MOV	R1,A	F9	1
MOV	R2,A	FA	1
MOV	R3,A	FB	1
MOV	R4,A	FC	1
MOV	R5,A	FD	1
MOV	R6,A	FE	1
MOV	R7,A	FF	1

MNEMONIC	OPERANDS	OPCODE	Cycles
MOV	R0,addr8	A8 aa	2
MOV	R1,addr8	A9 aa	2
MOV	R2,addr8	AA aa	2
MOV	R3,addr8	AB aa	2
MOV	R4,addr8	AC aa	2
MOV	R5,addr8	AD aa	2
MOV	R6,addr8	AE aa	2
MOV	R7,addr8	AF aa	2
MOVC	A,@A+DPTR	93	2
MOVC	A,@A+PC	83	2
MOVX	@DPTR,A	F0	2
MOVX	@R0,A	F2	2
MOVX	@R1,A	F3	2
MOVX	A,@DPTR	E0	2
MOVX	A,@R0	E2	2
MOVX	A,@R1	E3	2
MUL	AB	A4	4
NOP		00	1
ORL	A,#data	44 nn	1
ORL	A,@R0	46	1
ORL	A,@R1	47	1
ORL	A,addr8	45 aa	1
ORL	A,R0	48	1
ORL	A,R1	49	1
ORL	A,R2	4A	1
ORL	A,R3	4B	1
ORL	A,R4	4C	1
ORL	A,R5	4D	1
ORL	A,R6	4E	1
ORL	A,R7	4F	1
ORL	C,/bit	A0 bb	2
ORL	C,bit	72 bb	2
ORL	addr8,#data	43 aa nn	2
ORL	addr8,A	42 aa	1
POP	addr8	D0 aa	2
PUSH	addr8	C0 aa	2
RET		22	2
RETI		32	2
RL	A	23	1
RLC	A	33	1
RR	A	03	1
RRC	A	13	1
SETB	bit	D2 bb	1
SETB	C	D3	1
SJMP	rel	80 rr	2

MNEMONIC	OPERANDS	OPCODE	Cycles
SUBB	A,#data	94 nn	1
SUBB	A,@R0	96	1
SUBB	A,@R1	97	1
SUBB	A,addr8	95 aa	1
SUBB	A,R0	98	1
SUBB	A,R1	99	1
SUBB	A,R2	9A	1
SUBB	A,R3	9B	1
SUBB	A,R4	9C	1
SUBB	A,R5	9D	1
SUBB	A,R6	9E	1
SUBB	A,R7	9F	1
SWAP	A	C4	1
XCH	A,@R0	C6	1
XCH	A,@R1	C7	1
XCH	A,addr8	C5 aa	1
XCH	A,R0	C8	1
XCH	A,R1	C9	1
XCH	A,R2	CA	1
XCH	A,R3	CB	1
XCH	A,R4	CC	1
XCH	A,R5	CD	1
XCH	A,R6	CE	1
XCH	A,R7	CF	1
XCHD	A,@R0	D6	1
XCHD	A,@R1	D7	1
XRL	A,#data	64 nn	1
XRL	A,@R0	66	1
XRL	A,@R1	67	1
XRL	A,addr8	65 aa	1
XRL	A,R0	68	1
XRL	A,R1	69	1
XRL	A,R2	6A	1
XRL	A,R3	6B	1
XRL	A,R4	6C	1
XRL	A,R5	6D	1
XRL	A,R6	6E	1
XRL	A,R7	6F	1
XRL	addr8,#data	63 aa nn	2
XRL	addr8,A	62 aa	1

Remarks:

addr8 = 00H ~ 0FFH
addr11 = 0000H ~ 07FFH
addr16 = 0000H ~ 0FFFFH
rel = 00H ~ 0FFH
#data = #00H ~ #0FFH
bit = bit address = 00H ~ 0FFH

	FLAG				FLAG		
	C	OV	AC		C	OV	AC
ADD	X	X	X	SETB C	1		
ADDC	X	X	X	CLR C	0		
SUBB	X	X	X	CPL C	X		
MUL	0	X		ANL C, bit	X		
DIV	0	X		ANL C, /bit	X		
DA	X			ORL C, bit	X		
RRC	X			ORL C, /bit	X		
RLC	X			MOV C, bit	X		
				CJNE	X		

Table 11. Instruction Opcodes in Hexadecimal Order

Hex Code	Number of Bytes	Mnemonic	Operands
00	1	NOP	
01	2	AJMP	code addr
02	3	LJMP	code addr
03	1	RR	A
04	1	INC	A
05	2	INC	data addr
06	1	INC	@R0
07	1	INC	@R1
08	1	INC	R0
09	1	INC	R1
0A	1	INC	R2
0B	1	INC	R3
0C	1	INC	R4
0D	1	INC	R5
0E	1	INC	R6
0F	1	INC	R7
10	3	JBC	bit addr, code addr
11	2	ACALL	code addr
12	3	LCALL	code addr
13	1	RRC	A
14	1	DEC	A
15	2	DEC	data addr
16	1	DEC	@R0
17	1	DEC	@R1
18	1	DEC	R0
19	1	DEC	R1
1A	1	DEC	R2
1B	1	DEC	R3
1C	1	DEC	R4
1D	1	DEC	R5
1E	1	DEC	R6
1F	1	DEC	R7
20	3	JB	bit addr, code addr
21	2	AJMP	code addr
22	1	RET	
23	1	RL	A
24	2	ADD	A,#data
25	2	ADD	A,data addr
26	1	ADD	A,@R0
27	1	ADD	A,@R1
28	1	ADD	A,R0
29	1	ADD	A,R1
2A	1	ADD	A,R2
2B	1	ADD	A,R3
2C	1	ADD	A,R4
2D	1	ADD	A,R5
2E	1	ADD	A,R6
2F	1	ADD	A,R7
30	3	JNB	bit addr, code addr
31	2	ACALL	code addr
32	1	RETI	
33	1	RLC	A
34	2	ADDC	A,#data
35	2	ADDC	A,data addr
36	1	ADDC	A,@R0
37	1	ADDC	A,@R1
38	1	ADDC	A,R0
39	1	ADDC	A,R1
3A	1	ADDC	A,R2
3B	1	ADDC	A,R3
3C	1	ADDC	A,R4
3D	1	ADDC	A,R5
3E	1	ADDC	A,R6
3F	1	ADDC	A,R7
40	2	JC	code addr
41	2	AJMP	code addr
42	2	ORL	data addr,A
43	3	ORL	data addr,#data
44	2	ORL	A,#data
45	2	ORL	A,data addr
46	1	ORL	A,@R0
47	1	ORL	A,@R1
48	1	ORL	A,R0
49	1	ORL	A,R1
4A	1	ORL	A,R2
4B	1	ORL	A,R3
4C	1	ORL	A,R4
4D	1	ORL	A,R5
4E	1	ORL	A,R6
4F	1	ORL	A,R7
50	2	JNC	code addr
51	2	ACALL	code addr
52	2	ANL	data addr,A
53	3	ANL	data addr,#data
54	2	ANL	A,#data
55	2	ANL	A,data addr
56	1	ANL	A,@R0
57	1	ANL	A,@R1
58	1	ANL	A,R0
59	1	ANL	A,R1
5A	1	ANL	A,R2
5B	1	ANL	A,R3
5C	1	ANL	A,R4
5D	1	ANL	A,R5
5E	1	ANL	A,R6
5F	1	ANL	A,R7
60	2	JZ	code addr
61	2	AJMP	code addr
62	2	XRL	data addr,A
63	3	XRL	data addr,#data
64	2	XRL	A,#data
65	2	XRL	A,data addr

Table 11. Instruction Opcodes in Hexadecimal Order (Continued)

Hex Code	Number of Bytes	Mnemonic	Operands
66	1	XRL	A,@R0
67	1	XRL	A,@R1
68	1	XRL	A,R0
69	1	XRL	A,R1
6A	1	XRL	A,R2
6B	1	XRL	A,R3
6C	1	XRL	A,R4
6D	1	XRL	A,R5
6E	1	XRL	A,R6
6F	1	XRL	A,R7
70	2	JNZ	code addr
71	2	ACALL	code addr
72	2	ORL	C,bit addr
73	1	JMP	@A+DPTR
74	2	MOV	A,#data
75	3	MOV	data addr,#data
76	2	MOV	@R0,#data
77	2	MOV	@R1,#data
78	2	MOV	R0,#data
79	2	MOV	R1,#data
7A	2	MOV	R2,#data
7B	2	MOV	R3,#data
7C	2	MOV	R4,#data
7D	2	MOV	R5,#data
7E	2	MOV	R6,#data
7F	2	MOV	R7,#data
80	2	SJMP	code addr
81	2	AJMP	code addr
82	2	ANL	C,bit addr
83	1	MOVC	A,@A+PC
84	1	DIV	AB
85	3	MOV	data addr, data addr
86	2	MOV	data addr,@R0
87	2	MOV	data addr,@R1
88	2	MOV	data addr,R0
89	2	MOV	data addr,R1
8A	2	MOV	data addr,R2
8B	2	MOV	data addr,R3
8C	2	MOV	data addr,R4
8D	2	MOV	data addr,R5
8E	2	MOV	data addr,R6
8F	2	MOV	data addr,R7
90	3	MOV	DPTR,#data
91	2	ACALL	code addr
92	2	MOV	bit addr,C
93	1	MOVC	A,@A+DPTR
94	2	SUBB	A,#data
95	2	SUBB	A,data addr
96	1	SUBB	A,@R0
97	1	SUBB	A,@R1
98	1	SUBB	A,R0
99	1	SUBB	A,R1
9A	1	SUBB	A,R2
9B	1	SUBB	A,R3
9C	1	SUBB	A,R4
9D	1	SUBB	A,R5
9E	1	SUBB	A,R6
9F	1	SUBB	A,R7
A0	2	ORL	C,/bit addr
A1	2	AJMP	code addr
A2	2	MOV	C,bit addr
A3	1	INC	DPTR
A4	1	MUL	AB
A5		reserved	
A6	2	MOV	@R0,data addr
A7	2	MOV	@R1,data addr
A8	2	MOV	R0,data addr
A9	2	MOV	R1,data addr
AA	2	MOV	R2,data addr
AB	2	MOV	R3,data addr
AC	2	MOV	R4,data addr
AD	2	MOV	R5,data addr
AE	2	MOV	R6,data addr
AF	2	MOV	R7,data addr
B0	2	ANL	C,/bit addr
B1	2	ACALL	code addr
B2	2	CPL	bit addr
B3	1	CPL	C
B4	3	CJNE	A,#data,code addr
B5	3	CJNE	A,data addr,code addr
B6	3	CJNE	@R0,#data,code addr
B7	3	CJNE	@R1,#data,code addr
B8	3	CJNE	R0,#data,code addr
B9	3	CJNE	R1,#data,code addr
BA	3	CJNE	R2,#data,code addr
BB	3	CJNE	R3,#data,code addr
BC	3	CJNE	R4,#data,code addr
BD	3	CJNE	R5,#data,code addr
BE	3	CJNE	R6,#data,code addr
BF	3	CJNE	R7,#data,code addr
C0	2	PUSH	data addr
C1	2	AJMP	code addr
C2	2	CLR	bit addr
C3	1	CLR	C
C4	1	SWAP	A
C5	2	XCH	A,data addr
C6	1	XCH	A,@R0
C7	1	XCH	A,@R1
C8	1	XCH	A,R0
C9	1	XCH	A,R1
CA	1	XCH	A,R2
CB	1	XCH	A,R3

Table 11. Instruction Opcodes in Hexadecimal Order (Continued)

Hex Code	Number of Bytes	Mnemonic	Operands
CC	1	XCH	A,R4
CD	1	XCH	A,R5
CE	1	XCH	A,R6
CF	1	XCH	A,R7
D0	2	POP	data addr
D1	2	ACALL	code addr
D2	2	SETB	bit addr
D3	1	SETB	C
D4	1	DA	A
D5	3	DJNZ	data addr,code addr
D6	1	XCHD	A,@R0
D7	1	XCHD	A,@R1
D8	2	DJNZ	R0,code addr
D9	2	DJNZ	R1,code addr
DA	2	DJNZ	R2,code addr
DB	2	DJNZ	R3,code addr
DC	2	DJNZ	R4,code addr
DD	2	DJNZ	R5,code addr
DE	2	DJNZ	R6,code addr
DF	2	DJNZ	R7,code addr
E0	1	MOVX	A,@DPTR
E1	2	AJMP	code addr
E2	1	MOVX	A,@R0
E3	1	MOVX	A,@R1
E4	1	CLR	A
E5	2	MOV	A,data addr

Hex Code	Number of Bytes	Mnemonic	Operands
E6	1	MOV	A,@R0
E7	1	MOV	A,@R1
E8	1	MOV	A,R0
E9	1	MOV	A,R1
EA	1	MOV	A,R2
EB	1	MOV	A,R3
EC	1	MOV	A,R4
ED	1	MOV	A,R5
EE	1	MOV	A,R6
EF	1	MOV	A,R7
F0	1	MOVX	@DPTR,A
F1	2	ACALL	code addr
F2	1	MOVX	@R0,A
F3	1	MOVX	@R1,A
F4	1	CPL	A
F5	2	MOV	data addr,A
F6	1	MOV	@R0,A
F7	1	MOV	@R1,A
F8	1	MOV	R0,A
F9	1	MOV	R1,A
FA	1	MOV	R2,A
FB	1	MOV	R3,A
FC	1	MOV	R4,A
FD	1	MOV	R5,A
FE	1	MOV	R6,A
FF	1	MOV	R7,A